

## CYP1A2

Cytochrome P450 1A2

### Gene Function:

CYP1A2 encodes a liver enzyme that metabolizes various substances, most famously caffeine, but also some drugs and toxins. Genetic variations in this gene can cause individuals to be "fast" or "slow" caffeine metabolizers, which affects how long caffeine stays in the body.

| SNP ID   | EFFECT ALLELE | YOUR GENOTYPE | STATUS       |
|----------|---------------|---------------|--------------|
| rs762551 | C             | CA            | Heterozygous |

### Understanding your data:

Your genotype for **rs762551** is AC, which indicates that you are an intermediate metabolizer of caffeine.[\[ref\]](#) This variant is associated with a reduced risk of breast cancer in Caucasian women and in BRCA1 mutation carriers who drink coffee. It is also associated with a reduced risk of dementia in coffee drinkers. [\[ref\]](#)[\[ref\]](#)[\[ref\]](#)

### CYP1A2 Background Science:

CYP1A2 is an enzyme responsible for breaking down a wide range of substances, including caffeine, aflatoxin B1 from mold, endogenous substances (e.g., melatonin, bilirubin, estrogens), and several prescription medications. How effectively these substances are metabolized can be influenced by genetic variants of CYP1A2, categorizing people as either fast or slow metabolizers

CYP1A2 can also be influenced by external factors, with some substances acting as inducers (e.g., polycyclic aromatic hydrocarbons from cigarette smoke, cruciferous vegetables) that speed up the enzyme’s activity, while others act as inhibitors (e.g., curcumin, turmeric, peppermint, chamomile, dandelion, St. John’s wort), slowing it down.

Consequently, these factors can affect the clearance rate of substances, such as drugs, processed by CYP1A2.

The enzyme's speed has implications for health. For example, its rapid metabolism of pro-carcinogens into carcinogens can increase the risk of certain cancers, while slow or reduced function could potentially decrease the risk of liver toxicity from aflatoxin B. This speed can also affect how long the effects of certain substances, like caffeine, last in the body.

References: Please see the [full article on CYP1A2](#) with links to peer-reviewed studies.

*Disclaimer: This report is for informational and educational purposes only. It is not intended to treat, diagnose, or cure any condition.*

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